

Apache Airflow is an **open-source workflow orchestration tool**.

In simple words: **Airflow is used to schedule, run, monitor, and manage data pipelines or automation workflows.**

Example:

You want to run this every day:

1. Read data from a CSV/API/database
2. Clean the data
3. Load it into a database or data lake
4. Run a report
5. Send an email notification

Apache Airflow helps you automate this entire flow.

Core idea

Airflow workflows are written as **DAGs**.

DAG = Directed Acyclic Graph

A DAG means a set of tasks connected in a fixed order.

Example:

```
extract_data → clean_data → load_data → send_email
```

Each box is a **task**.

Why companies use Airflow

Airflow is useful when you have jobs that must run:

- Daily, hourly, weekly, or at a fixed time
- In a specific order
- With retry logic if something fails
- With monitoring and logs
- Across multiple systems like databases, APIs, cloud storage, Spark, Databricks, etc.

Simple real-world example

Suppose a company wants a sales report every morning.

Without Airflow:

Someone manually downloads sales data, cleans it, uploads it, and runs reports.

With Airflow:

```
Every day at 6 AM:
```

1. Download sales data
2. Validate records
3. Clean null values
4. Load into warehouse
5. Refresh dashboard
6. Send success/failure email

Airflow does this automatically.

Important Airflow components

Component	Meaning
DAG	Complete workflow
Task	One step inside the workflow
Operator	Type of task, such as Python task, Bash task, SQL task
Scheduler	Decides when DAGs should run
Executor	Runs the tasks
Web UI	Browser interface to monitor DAGs
Metadata DB	Stores task status, logs, schedule details

Tiny Airflow DAG example

```
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime
```

```
def say_hello():  
    print("Hello from Airflow")  
  
with DAG(  
    dag_id="hello_airflow_dag",  
    start_date=datetime(2024, 1, 1),  
    schedule="@daily",  
    catchup=False  
) as dag:  
  
    task1 = PythonOperator(  
        task_id="say_hello_task",  
        python_callable=say_hello  
    )
```

This creates a DAG that runs daily and prints:

```
Hello from Airflow
```

Airflow is mainly used for

- Data engineering pipelines
- ETL / ELT workflows
- Data warehouse loading
- Machine learning pipeline scheduling
- Report automation
- Cloud job orchestration
- Spark / Databricks / Python / SQL job scheduling

In one line

Apache Airflow is like a smart scheduler and controller for your data workflows.